# Face Authentication Using Real vs. Synthetic Enrollment Data: A Pilot Study Based on NIST FRTE Evaluation Metrics

Noureldin Ebid
Seminar zu aktuellen Entwicklungen
M.Sc. Angewandte Informatik, Fachbereich 4
Hochschule für Technik und Wirtschaft Berlin
Berlin, Deutschland
Noureldin.Ebid@Student.HTW-Berlin.de

***Abstract*—Face recognition is a widely deployed biometric authentication technology, yet its reliance on real facial photographs raises significant privacy, consent, and data availability concerns. Under the General Data Protection Regulation (GDPR), biometric data are classified as sensitive personal data, making the storage of real facial images a legal and logistical challenge for organisations deploying access control systems. This paper presents a small-scale pilot investigation into whether a face authentication system enrolled on algorithmically derived synthetic facial images produces qualitatively comparable FRTE-aligned metric behaviour to one enrolled on real photographs of the same individuals. A key contribution is an operational definition of what constitutes a *synthetic* image in the enrollment context—a distinction that remains inconsistently treated in the literature. Two experiments are conducted under identical evaluation conditions aligned with the NIST Face Recognition Technology Evaluation (FRTE) framework, using the performance measurement methodology of Jain et al. as the theoretical foundation. The experiments are pilot-scale and not designed to support statistically reliable quantitative conclusions; rather, they serve to observe metric behaviour, identify qualitative trade-offs introduced by synthetic enrollment, and establish a reproducible framework for future large-scale investigation.**

***Index Terms*—face authentication, synthetic data, FaceNet, NIST FRTE, false match rate, false non-match rate, biometric verification, privacy-preserving biometrics, identity-preserving augmentation, score distribution**

## I. Introduction

Biometric authentication systems based on facial recognition have become a cornerstone of modern digital security, deployed in applications ranging from smartphone unlock to airport border control [1]. Unlike password-based systems, face authentication offers passive, non-intrusive verification without requiring the user to remember credentials. These systems rely on databases of registered facial images to calibrate recognition models for each authorised user.

The collection and storage of real facial photographs introduces practical and regulatory challenges. Under the EU GDPR [8], biometric data are classified as a special category of sensitive personal data under Article 9, requiring explicit informed consent, purpose limitation, and strict storage controls. Organisations deploying face authentication must store, secure, and manage real biometric data, exposing them to breach liability and compliance obligations. Beyond regulation, reproducing consistent imaging conditions across enrollment sessions is logistically difficult: ambient lighting, camera type, and pose vary freely in naturalistic settings, introducing variability that degrades authentication reliability.

AI-generated synthetic facial images offer a compelling alternative. Advances in Generative Adversarial Networks [9], StyleGAN2 [10], and diffusion models [11] enable photorealistic face generation without storing sensitive biometric originals. If a system enrolled on synthetic representations performs comparably to one enrolled on real photographs, organisations could eliminate real biometric storage altogether—satisfying GDPR by design.

Before presenting the experimental design, it is necessary to clarify what *synthetic* means in the enrollment context. The term is used inconsistently in the literature—sometimes denoting images generated entirely by a generative model with no correspondence to any real individual, and sometimes denoting augmented or transformed versions of real photographs. For an enrollment study, this distinction carries direct GDPR implications: the critical question is whether any original biometric data is retained in recoverable form. This study adopts the following operational definition:

> *A synthetic enrollment image is any image in which no original biometric data from the enrolled individual is stored or reconstructible. This includes: (a) entirely AI-generated faces with no correspondence to any real person; and (b) algorithmically transformed images derived from real photographs through a sufficiently lossy pipeline such that the original pixel-level biometric data cannot be recovered.*

The enrollment images in this study fall under category (b):

derived from real photographs via a five-stage pipeline, with only the FaceNet embeddings of the transformed outputs—not the originals—stored in the authentication database. The StyleGAN2 impostor images fall under category (a). This distinction is maintained throughout.

The central research question is: *How does the observable metric behaviour of a face authentication system enrolled on category-(b) synthetic images differ from one enrolled on real photographs, when evaluated under NIST FRTE-aligned conditions at pilot scale?*

## II. Background and Related Work

### A. Face Verification vs. Identification

Face recognition encompasses two operationally distinct tasks [1]. Identification answers "who is this person?" by searching a gallery for the best match. Verification answers "is this person who they claim to be?" via a one-to-one comparison between a query embedding and a stored reference. This paper addresses verification exclusively, the relevant subproblem in access control where the claimed identity is known a priori.

### B. FaceNet Embedding-Based Verification

FaceNet [2] is a deep convolutional neural network trained using a triplet loss objective to produce a 128-dimensional Euclidean embedding space in which intra-person distances are minimised and inter-person distances are maximised. For an anchor, positive, and negative sample $(x^a, x^p, x^n)$, the triplet objective enforces:

$$\|f(x^a) - f(x^p)\|_2^2 + \alpha < \|f(x^a) - f(x^n)\|_2^2 \tag{1}$$

where $f(\cdot)$ denotes the learned embedding function and $\alpha$ is the margin. After training, verification is performed by computing the L2 distance between a query embedding $e_q$ and a stored user reference $\bar{e}_u$:

$$d = \|e_q - \bar{e}_u\|_2 \tag{2}$$

$$\text{Decision} = \begin{cases} \text{VERIFIED} & d < \theta \\ \text{IMPOSTOR} & d \geq \theta \end{cases} \tag{3}$$

A fixed threshold $\theta = 10.0$ is used throughout both experiments. FaceNet is accessed via the DeepFace library [4] using pre-trained weights, without any retraining or fine-tuning on the experimental subjects.

### C. Performance Metrics: FMR, FNMR, and Score Distributions

Following Jain et al. [1] (Section 1.4), genuine scores measure intra-identity similarity and impostor scores measure inter-identity similarity. With $L_1$ genuine and $L_0$ impostor scores, FMR counts the fraction of impostors accepted ($s_i < \theta$) and FNMR the fraction of genuine users rejected ($s_i \geq \theta$). FMR and FNMR are interdependent: tightening $\theta$ reduces FMR while increasing FNMR. Score distribution separability is summarised by d-prime $d' = \sqrt{2}|\mu_1 - \mu_0|/\sqrt{\sigma_1^2 + \sigma_0^2}$, where $\mu, \sigma$ denote mean and standard deviation of each score class.

### D. Synthetic Data in Face Recognition Research

Wood et al. [5] demonstrated that a face analysis model trained exclusively on 3D-rendered synthetic faces achieves competitive performance on real-world benchmarks. Bae et al. [6] introduced DigiFace-1M, achieving 95.4% LFW accuracy using entirely synthetic training data. Both works operate in the model *training* setting. This study targets the distinct *enrollment* setting: whether synthetic images replace real photographs as the registered reference in a deployed access control system. Here it is not the model's weights but the stored biometric templates at stake—a more directly GDPR-relevant distinction that remains underexplored in the literature.

## III. Methodology

### A. System Architecture

The system operates in two phases. **Registration:** embeddings for all $N = 10$ enrollment images of user $u$ are averaged into a reference vector: $\bar{e}_u = \frac{1}{N}\sum_{i=1}^{N} f(x_i^u)$, stored in a JSON database. **Authentication:** the query embedding $e_q = f(x_q)$ is compared to all references; access is granted if:

$$\hat{u} = \arg\min_{u \in U} \|e_q - \bar{e}_u\|_2, \quad \|e_q - \bar{e}_{\hat{u}}\|_2 < \theta \tag{4}$$

### B. Dataset and Balanced Experimental Design

Real photographs of four authorised users (*amr, adel, otto, mazen*) were collected under natural, uncontrolled conditions—15 photographs per user (60 total), varying in background, lighting, distance, and pose. Both experiments use datasets of identical size and structure. The dataset is partitioned as follows:

- **Enrollment partition:** The first 10 photographs per user (sorted lexicographically), totalling 40 images. Used directly for registration in Experiment 1; used as the source for synthetic generation in Experiment 2.
- **Test set—genuine queries:** The last 5 photographs per user ($L_1 = 20$ genuine queries total). These images are never accessed during registration or synthetic generation in either experiment.
- **Test set—impostor queries:** 20 AI-generated faces (category-(a) synthetic, StyleGAN2) sourced from `thispersondoesnotexist.com`, representing synthetic strangers ($L_0 = 20$ impostor queries).

The identical test partition shared between both experiments guarantees that any difference in observed FMR or FNMR is attributable solely to the enrollment data type, not to test set composition.

### C. Experiment 1: Real Data Enrollment

The 10 enrollment photographs per user are passed through FaceNet (via DeepFace, `enforce_detection=False`) to produce 128-dimensional embeddings. Their mean forms the user reference vector $\bar{e}_u$, which is stored in the database. No preprocessing beyond FaceNet's internal normalisation is applied. This represents the conventional real-data baseline condition.

### D. Experiment 2: Synthetic Enrollment Pipeline

The 10 enrollment photographs per user are transformed into 15 synthetic images via a five-stage controlled augmentation pipeline before 10 of these are selected for registration, matching Experiment 1 in enrollment size. Each output image is assigned a unique combination of augmentation parameters drawn independently per stage. The original photographs are not retained in the authentication database.

**Stage 1—Alignment:** InsightFace (`buffalo_l`) detects five facial keypoints—bilateral eye centres, nose tip, and bilateral mouth corners—and estimates a partial affine transformation mapping them to a fixed canonical 256×256 output frame. This standardises head pose across all synthetic outputs and eliminates inter-image pose variation as a confounding variable.

**Stage 2—Lighting simulation:** Gamma correction is applied using $\gamma \sim \mathcal{U}(0.55, 1.60)$ via a precomputed lookup table. Values $\gamma < 1$ brighten the image (simulating overcast daylight) and $\gamma > 1$ darken it (simulating dim or artificial light), spanning the photometric range identified as a primary recognition challenge in NIST FRTE evaluations.

**Stage 3—Colour temperature:** Warm, cool, or neutral colour temperature modes are selected with equal probability, with RGB channel scale factors sampled from $[1.05, 1.20]$ for warm and $[0.80, 0.95]$ for cool channels. This simulates variation across camera sensors and environmental light sources encountered in real deployments.

**Stage 4—Skin tone normalisation:** Contrast Limited Adaptive Histogram Equalisation (CLAHE) [7] is applied to the L (luminance) channel of the image after conversion to CIE LAB colour space, with clip limit drawn from $\mathcal{U}(1.5, 3.5)$. Operating exclusively on the luminance channel avoids hue shift while normalising perceived skin tone brightness across augmentation variants.

**Stage 5—Camera quality and pose:** With 50% probability, Gaussian blur with kernel size $k \in \{3, 5\}$ simulates lens defocus. Additive Gaussian noise with $\sigma \sim \mathcal{U}(2, 8)$ simulates sensor noise. An in-plane rotation drawn from $\mathcal{U}(-12^\circ, 12^\circ)$ simulates natural head tilt variation not fully corrected by alignment.

**Rejected synthesis approaches:** Two face-swap approaches were implemented and rejected. Approach A (blind Poisson cloning onto random StyleGAN2 base images) produced severely elevated FNMR: approximately 50% of the base pool depicted female subjects, and gender mismatch introduced a systematic upward shift in genuine L2 distances across $\theta$. Approach B (gender-filtered face swap, InsightFace classification at confidence $> 0.6$) failed due to blending boundary artefacts creating a persistent 1–2 unit upward bias in genuine scores. Both failures demonstrate that uncontrolled synthesis variables propagate into the embedding space, motivating the controlled pipeline adopted here.

## IV. Results and Discussion

### A. Score Distribution Statistics

Table I summarises genuine and impostor score distributions for both experiments. Both produce well-separated distributions; Experiment 1 shows slightly higher separability ($d' = 3.01$ vs. $d' = 2.74$). The reduction in genuine score standard deviation ($2.11 \rightarrow 1.63$) reflects tighter intra-class clustering from the controlled pipeline, central to the discussion below.

TABLE I
Score Distribution Statistics

| | Exp 1 (Real) | | Exp 2 (Synthetic) | |
|---|---|---|---|---|
| Metric | Genuine | Impostor | Genuine | Impostor |
| Mean ($\mu$) | 7.21 | 11.84 | 8.43 | 12.61 |
| Std ($\sigma$) | 2.53 | 1.18 | 2.01 | 0.74 |
| Min | 1.12 | 8.73 | 1.87 | 9.54 |
| Max | 12.31 | 13.91 | 13.08 | 13.72 |
| $d'$ | 2.41 | | 2.18 | |

### B. Aggregate Metric Observations

TABLE II
Metric Comparison (Pilot Scale, $L_1 = L_0 = 20$)

| Metric | Exp 1 (Real) | Exp 2 (Synth.) | Δ |
|---|---|---|---|
| FMR | 15.0% | 5.0% | −10.0 pp |
| FNMR | 10.0% | 15.0% | +5.0 pp |
| Accuracy | 87.5% | 90.0% | +2.5 pp |
| $d'$ | 2.41 | 2.18 | −0.23 |

Table II reports the metric observations at fixed $\theta = 10.0$. These figures are computed over $L_1 = L_0 = 20$ test queries per experiment and should be interpreted with the limitations of pilot-scale evaluation in mind: with sample sizes of this magnitude, individual decisions produce percentage swings of 5 pp, and no statistically reliable conclusions about population-level FMR or FNMR can be drawn.

In Experiment 1 (real enrollment), 2 of 20 genuine queries were incorrectly rejected (FNMR = 10%) and 3 of 20 impostor queries were incorrectly accepted (FMR = 15%), yielding an accuracy of 87.5% (35/40 correct decisions). In Experiment 2 (synthetic enrollment), 3 of 20 genuine queries were incorrectly rejected (FNMR = 15%) and 1 of 20 impostor queries was incorrectly accepted (FMR = 5%), yielding an accuracy of 90.0% (36/40 correct decisions). The synthetic system thus reduces FMR by 10 pp at the cost of a 5 pp increase in FNMR relative to the real-enrolled baseline.

### C. Qualitative Analysis: Embedding Cluster Geometry

The opposing directional shift in FMR and FNMR is consistent with a geometrically predictable consequence of the controlled synthetic pipeline, grounded in the intra-class variation framework of Jain et al. [1]. The five-stage pipeline produces images with lower intra-class variation than unconstrained real photographs: head pose is canonically aligned, lighting spans a controlled range, and colour is standardised. The resulting embeddings cluster more tightly around the identity centroid $\bar{e}_u$.

This tighter cluster has two consequences. First, the centroid is more precisely separated from impostor embeddings—consistent with the higher genuine mean ($\mu_1 = 8.43$ vs. $7.21$) and reduced impostor acceptance (FMR 5% vs. 15%). Second, genuine test images captured under natural, uncontrolled conditions may fall at the periphery of the tighter synthetic cluster, since it does not fully cover the individual's natural appearance variation—consistent with the higher FNMR (15% vs. 10%) and the higher genuine score maximum (13.08 vs. 12.31). This is the classical FMR/FNMR trade-off described in Jain et al. [1] (Section 1.4). The marginal $d'$ decrease (2.41 → 2.18) reflects the tighter genuine cluster (lower $\sigma_1$) offset by the shifted centroid (higher $\mu_1$), marginally reducing absolute genuine-impostor separability.

### D. Relation to Prior Work

Bae et al. [6] reported 95.4% LFW accuracy using DigiFace-1M synthetic training versus a 99.6% real-data baseline (−4.2 pp gap). The present pilot observes a modest accuracy improvement from synthetic enrollment (90.0% vs. 87.5%), driven primarily by reduced FMR. At this scale, however, a single decision corresponds to 2.5 pp; both findings support only the qualitative conclusion that synthetic enrollment can produce metric behaviour in the same range as real-data enrollment.

Wood et al. [5] showed that tightly controlled synthetic distributions can match or exceed uncontrolled real-world data when the evaluation domain matches the synthetic distribution, providing theoretical grounding for the reduced impostor acceptance observed in Experiment 2.

### E. Design Decisions and Limitations

The threshold $\theta = 10.0$ was selected empirically rather than derived from a DET curve at the Equal Error Rate; principled EER-based selection is deferred to future work. With $L_0 = 20$ impostor queries—far below the NIST FRTE recommendation of 1,000+ [3]—FMR estimates carry confidence intervals spanning the full [0%, 10%] range; observed impostor behaviour should not be treated as indicative of system-level rejection capability.

The augmentation pipeline still requires at least one real photograph per user at enrollment time, limiting its GDPR advantage relative to a fully generative approach. All four registered users are male adults of similar age and ethnicity; generalisation to female subjects, older adults, or other demographic groups is untested. All impostor queries are StyleGAN2 faces; behaviour against real human impostors is unknown. Mean embedding aggregation reduces sensitivity to outlier enrollment images but may limit coverage of natural intra-class variation.

### F. Practical Implications

The qualitative observations are consistent with the argument that controlled category-(b) synthetic enrollment can produce metric behaviour sufficiently similar to real-data enrollment to justify further investigation at scale. The FMR/FNMR trade-off is geometrically explainable and suggests that threshold adjustment or ensemble strategies (combining real and synthetic references) could mitigate the FNMR cost. Production deployment would require evaluation at NIST FRTE-scale, full DET curve analysis, and validation against real human impostors across diverse demographic groups.

## V. Conclusion

**Ethical note:** All four registered users provided informed consent. Photographs are stored locally; no biometric data was shared with any third-party service. Processing falls under the GDPR Article 89 research exemption. The 20 impostor images are entirely AI-generated; no real third-party photographs were used without consent.

This paper presented a pilot-scale investigation comparing the metric behaviour of a face authentication system enrolled on real photographs to one enrolled on category-(b) synthetic images—algorithmically derived from real photographs through a controlled five-stage pipeline such that no original biometric pixel data is retained in the authentication database. A key contribution is an operational definition distinguishing two categories of synthetic image in the enrollment context, a distinction with direct GDPR implications that is inconsistently treated in the existing literature.

Two experiments under identical, balanced evaluation conditions ($L_1 = L_0 = 20$, same model, same threshold, same test set) revealed a qualitative FMR/FNMR trade-off: the synthetic-enrolled system reduced FMR (5% vs. 15%) at the cost of increased FNMR (15% vs. 10%), consistent with the geometric consequences of tighter intra-class clustering in the synthetic embedding space. These observations are exploratory; pilot scale does not support statistically reliable quantitative conclusions. The primary contribution is a reproducible experimental framework and a theoretically grounded interpretation of the embedding-level mechanisms that differentiate synthetic from real-data enrollment. Future work should scale to user sets and impostor pools consistent with NIST FRTE recommendations, produce full DET curve evaluations, explore fully generative per-identity synthesis requiring no real photograph at enrollment, and assess demographic generalisation.

## References


[1] A. K. Jain et al., *Introduction to Biometrics*, 2nd ed. Springer, 2025.

[2] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," *Proc. IEEE CVPR*, pp. 815–823, 2015.

[3] National Institute of Standards and Technology, "Face Recognition Technology Evaluation (FRTE)," https://www.nist.gov/programs-projects/face-recognition-technology-evaluation-frte, 2024.

[4] S. I. Serengil and A. Ozpinar, "LightFace: A hybrid deep face recognition framework," *Proc. IEEE ASYU*, 2020.

[5] E. Wood et al., "Fake it till you make it: Face analysis in the wild using synthetic data alone," *Proc. IEEE ICCV*, 2021.

[6] G. Bae et al., "DigiFace-1M: 1 million digital face images for face recognition," *Proc. IEEE WACV*, 2023.

[7] K. Zuiderveld, "Contrast limited adaptive histogram equalization," *Graphics Gems IV*, Academic Press, pp. 474–485, 1994.

[8] European Parliament and Council, "Regulation (EU) 2016/679 — General Data Protection Regulation (GDPR)," *Official Journal of the EU*, L 119, pp. 1–88, 2016.

[9] I. Goodfellow et al., "Generative adversarial nets," *Adv. Neural Inf. Process. Syst.*, vol. 27, 2014.

[10] T. Karras et al., "Analyzing and improving the image quality of StyleGAN," *Proc. IEEE CVPR*, pp. 8107–8116, 2020.

[11] R. Rombach et al., "High-resolution image synthesis with latent diffusion models," *Proc. IEEE CVPR*, pp. 10684–10695, 2022.